

Isaiah Salcedo

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Seeking to apply my knowledge of power systems and circuit theory in an Electrical Engineering internship.

EDUCATION

Purdue University

Anticipated May 2027

- Bachelor of Science in Electrical Engineering, GPA: 3.1/4.0

EXPERIENCE

President | Latinos in Science and Engineering (MAES)

August 2023 - Present

- Spearheaded a membership growth initiative that **quadrupled club size** from 10 to over 40 members in one semester by fostering a collaborative environment.
- Organized and led 12 professional development workshops and outreach events, demonstrating strong **written and verbal communication skills**.
- Managed all club operations, including budget allocation and project milestones, exhibiting excellent **time management** and ability to balance multiple tasks.

Avionics Sub-Team Member | Purdue Orbital

August 2024 - Present

- Applied fundamental concepts of **power distribution** and protection to design and validate avionics power systems, performing **AC and DC circuit analysis** using **LTspice**.
- Demonstrated strong **analytical thinking** by interpreting complex data from oscilloscopes to troubleshoot and develop engineering solutions for power quality issues.
- Contributed to a **multidisciplinary group**, collaborating with mechanical and systems teams to ensure proper integration of power and data subsystems.

Electrical Engineering Intern | ASTECH Engineered Products

May 2025 - August 2025

- Assisted in the schematic design of **power distribution boards**, applying precise calculations and modeling with high **attention to detail**.
- Authored and maintained comprehensive **engineering documentation**, including technical drawings and test procedures, to present technical information clearly.

Full-Cycle PCB Design | Custom LED Array Personal Project

February 2026

- Managed the complete design of a custom DC-powered LED array, from schematic capture in **KiCad** to generating Gerber files for fabrication.
- Gained hands-on **lab experience** by performing precision soldering of mixed SMD/THT components and conducting power-on testing to validate the final assembly.

Chassis Robotics Sub-Team Member | Boiler Robotics

January 2024 - May 2024

- Developed mechanical prototypes using **3D Printing** for rapid iteration of custom components and utilized **MATLAB** for performance modeling of the chassis system.
- Exhibited a collaborative **team orientation** by working with other engineering disciplines to integrate the chassis with power and control systems for a modular robotic platform.

Suspension Sub-Team Member | Purdue Baja Racing

August 2024 - May 2025

- Assured vehicle safety and effectiveness by directing the sub-team through **CAD modeling** of the suspension system using **SolidWorks**.
- Ensured competition readiness by manufacturing and **testing** the final suspension assembly, which successfully passed all technical inspections.

SKILLS

Power Distribution, Power Quality, Circuit Protection, KiCad, LTspice, MATLAB, C/C++, Python, Technical Documentation, Analytical Thinking, Data Interpretation, Microsoft Office Suite (Excel, Word, Teams)

COURSEWORK

Power Systems Analysis, AC/DC Circuit Theory, Electronics, Electromagnetics, Three-Phase Systems, C Programming